IMY 120

PROJECT 4: THEME 5

🔎 OVERVIEW

This mini-project is based on the skills that you have learnt in theme 5. This project requires you to create a mini endless runner game, similar to the Google Chrome dinosaur game shown when a user does not have an internet connection (http://www.trex-game.skipser.com/).

Your game does not need to be as polished as the Google Chrome game, but the concepts are similar. The user presses the spacebar to make the character jump over the obstacles. Over time, the obstacles appear more quickly. The game ends when the character hits one of the obstacles.

A full example of what your project should be (excluding graphics) can be found on clickUP in the Project 4 folder. Take note that this example uses placeholder graphics – your game should have nice-looking visuals.

You do not need to show progress for this project.

🌟 PURPOSE

This project helps you apply what you've learned in Theme 5 along with all the previous themes by creating a fully functional game, including graphics and sound.

📌 REQUIREMENTS

Your project must meet the following requirements. Use the provided example video to guide you.

General document settings

- Your stage size should be 1280x720 pixels.
- The frame rate should be 24 frames per second.
- Your timeline should contain only a single frame with a script attached.
- All assets must be loaded dynamically.
- Everything except the running animation must be done using JavaScript. This includes the jumping.

Loading the game

- The game should start by loading all the assets in their correct positions – character, floor, obstacles and start button. When the start button is pressed, the game should start.

Jumping

- The player must press **spacebar** to make the character jump. You may not use another key for this.
- The player should not be able to jump while they are currently jumping.
- Player jump – to receive a passing mark for this section (50%) the jump should be uniform speed (same speed up and down with no pause at the top). If you implement a jump with acceleration, deceleration and a pause at the top, then you will receive more marks for this section. The example of the game above shows a jump with acceleration, deceleration and a pause at the top. **The jump must be done using JavaScript, not an internal animation.**
- The character should jump to the **same height** every time, this should not be variable.
- When the character jumps, the running animation should pause.
- The player jump must have an appropriate sound effect. You can find free sound effects at: https://freesound.org/

Score

- If the player jumps over an obstacle successfully, their score should increase by **10 points** and this must be displayed in the **top-right corner** of the screen.
- The game **speed** should increase every **100 points**. You can decide by how much the speed increases but be careful not to make it too difficult too quickly.

Obstacles

- Obstacles must be **randomly generated** in terms of their **distance** from each other. Make sure to keep this distance within a logical range – i.e. not too close to making jumping two obstacles impossible and not too far to make the wait between jumps too long and boring. The game should be able to continue **forever**, meaning you need to continue generating obstacles for as long as necessary. Make sure to remove them from the screen when they are no longer visible, or the browser will run out of memory.
- If the character hits an obstacle – whilst jumping or running – there should be a slight pause and then the game should reset. A sound effect should also accompany this collision event.

Visual assets

- **Visual assets** – you should choose/create/download your own visual assets for the game. You can choose any theme for your assets. You may use **freely** available vector images, but not copyrighted images. You will need the following visual assets:
- **A character** – this is a simple character with a basic walking/running animation as an internal animation (3-4 frames). **This is the only internal animation allowed.**
- **Obstacles** – you only need one obstacle asset that can be reused for all the obstacles in the game
- **Ground plane** – this is the ground on which the character runs – it can be a simple straight line or something else

- **Start button** – this is the button that the player presses to start the game
- **Score indicator** – this is the textbox in the top-right corner which shows the user their current score. Remember to submit any fonts used.
- Note: your assets should all make sense together. For example: an underwater theme (underwater background) with shells as obstacles and a character who is a scuba diver. Another example: a Western theme, with a cowboy character with beer bottles as obstacles.

Tips

- Don't make the character move across the stage, make the obstacles move towards the character, who remains in one place.
- Explore the use of JavaScript arrays for keeping track of the obstacles (https://www.w3schools.com/js/js_arrays.asp)
- Explore the use of the timeout function for inserting pauses before events take place (https://www.w3schools.com/js/js_timing.asp)
- Don't have too many obstacles on the stage at a time (whether visible or not) or the browser will run out of memory. Rather recycle your assets by adding and removing them as needed.
- Start working on this project early otherwise you will not have enough time to solve all the logic problems.

📅 DEADLINES

This project has one deadline only since feedback is not required for a project of this nature. Please see the module schedule for submission dates.

📦 SUBMISSION INSTRUCTIONS

- Read the specification again to make sure you have done everything.
- Read the rubric again to make sure you have done the best that you can.

Submit the following files in a zip folder:

- Your Adobe Animate file named Surname_Project4.fla (where surname is your surname). Do not submit your published game files.
- A folder containing any sound assets that you are using in the game. Make sure that the file path to these assets is correct in your code, otherwise your sound will not play properly.
- Any font files that you have used.

Name the zip folder Surname_Project4.zip (where surname is your surname). Submit the zip folder on clickUP before the deadline. Once you have submitted, download your submission and run it to make sure that everything works.

⚠️ IMPORTANT NOTES

- Your game will be marked by being opened in Animate and then published using Ctrl+Enter. It will be marked in the latest version of Firefox.
- If some code is preventing the game from running, comment it out. We will not mark your code, only the parts of the game that run.

📊 MARKING RUBRIC

	NOT ACHIEVED	PARTIALLY ACHIEVED	ACHIEVED	EXCEEDS EXPECTATIONS
VISUALS	**0**	**1**	**2**	**3**
	Placeholder visuals used.	Very little effort has gone into making the visuals attractive.	Acceptable visuals.	Very attractive visuals which enhance the feeling of the game.
JUMP	**0**	**1-2**	**3**	**4**
	Jump does not work/jump is implanted as an internal animation/does not work with spacebar.	Jump is buggy/sometimes ends below floor level (1) Jump is acceptable (uniform speed)/jump has variable heights (2)	Jump has acceleration/ deceleration/a pause at the top, but it needs some more tweaking to make it good.	Excellent jump – with acceleration, deceleration and a pause at the top.
OBSTACLES	**0**	**1**	**2**	**4**
	No obstacles.	Obstacles are buggy/are not randomly generated/the game is not endless (obstacles stop)	Obstacles work, but the distances between them should be tweaked. Game is endless.	Obstacles are well-spaced and recycled. Game is endless.
SOUND EFFECTS	**0**	**1**	**4**	
	No sound added/sound does not work.	Only one sound effect added (jump or hit) Both sounds added but the sound is buggy.	Applicable sound effects and both added.	
CHARACTER	**0**	**2**	**3**	
	No animation.	Character has running animation applied, but this is buggy.	Animation correctly applied and works at all times. Stops during jumping.	
COLLISION	**0**	**1-2**	**3**	
	Collision does not work	Collision works but is too sensitive at times.	Collision works.	
GAME START AND RESTART	**0**	**1**	**2**	**3**
	Game can't start or restart.	Game can start but not restart.	Game can start and restart, but it has bugs upon restarting.	Game starts and restarts perfectly. It is endlessly playable.
SCORE AND DIFFICULTY	**0**	**1**	**3**	
	Score not shown/shown but not updated.	Score shown and updated. No difficulty increase every 100 points.	Score updates and difficulty increases every 100 points.	